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Late Permian palaeomagnetic results from the Brive basin (Massif Central, France)

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Abstract

A palaeomagnetic study has been carried out in the Brive basin in order to understand the continental deformation of the French Massif Central since the Permian. Fifteen sites of Permian-age red sandstones have been sampled in the Brive basin which is situated in the west of the Sillon Houiller fault (France). Laboratory analyses have revealed well-behaved and only reversed magnetic polarity for the higher-temperature component. The effects of positive folding test and Kiaman reversal chron indicate that the magnetic remanent is probably primary. The so-called 'Autunian' and 'Saxo-Thuringian' stages show very close palaeomagnetic directions, with an angular difference of $1.2 \pm 6.9°$. They likely correspond to the same geological period (i.e. Late Permian), and allow us to calculate one palaeomagnetic pole for this basin (48.8°N, 162.5°E, $n = 15$ and $A_{95} = 2.6°$). Comparing this result with previous palaeomagnetic studies from other Permian basins in the south Massif Central and Stable Europe, we may conclude that: (1) the Brive basin has been part of Stable Europe since, at least, the Late Permian, and no younger palaeomagnetically detectable movement of this basin has be found with respect to Stable Europe; (2) no significant movement along the Sillon Houiller fault has been detected since the Permian. The relative latitudinal movement of the Permian basins of the Massif Central (i.e. Brive, Saint-Affrique and Lodève) is negligible. No significant rotation for each of these basins with respect to Stable Europe has been observed, except for the St. Affrique basin. This later basin experienced a counter-clockwise rotation of ~7° with respect to Stable Europe which represents probably a local phenomenon.

Keywords: palaeomagnetism; Massif Central; Permian; Brive; Stable Europe

1. Introduction

The Variscan orogen of western Europe is a collisional belt between the northern Laurussia and southern Gondwana continents with an age span from Late Silurian to Late Carboniferous–Early Permian (e.g., Matte, 1986; Dewey and Burke, 1988). The French Massif Central is one of the key areas to understand the kinematics of Gondwana with respect to Laurussia. There, the late stages of the collision, since Middle Carboniferous, are characterized by extensional tectonics. Syntectonic plutons emplacement, metamorphic retrogression and intramontane sedimentary basins are conspicuous. It has been shown that the development of these late orogenic extensional structures is controlled by two successive trends. Namely, in Middle Carboniferous a NW–SE

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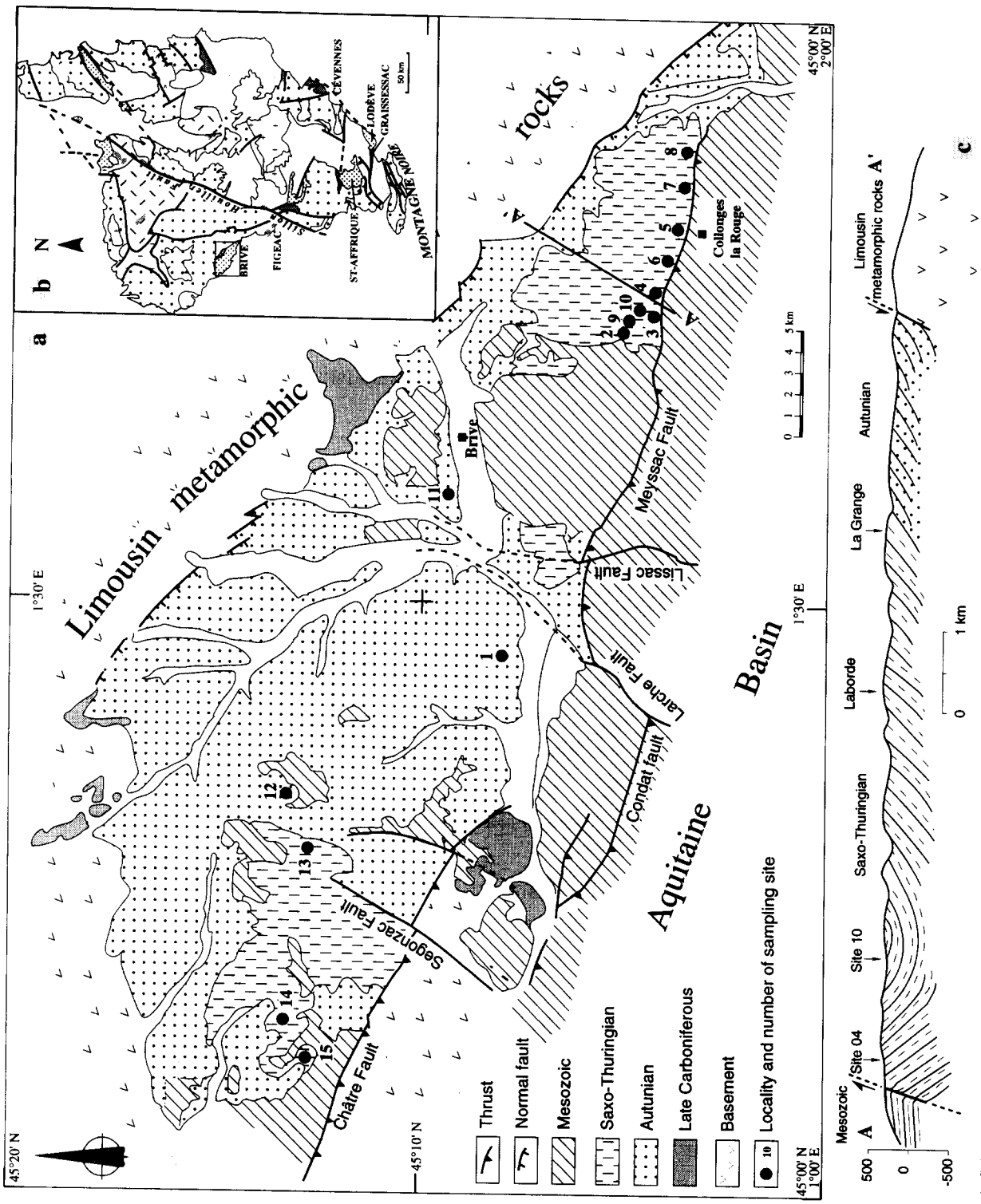


Fig. 1. Schematic tectonic and geological maps for the French Massif Central (b), the Brive Basin (a) (according to Feys, 1989) and a geological cross-section of sampling area (according to Lefavrais-Raymond and Feys, 1976).

direction and in Late Carboniferous–Early Permian the NE–SW trend is dominant (Faure, 1995). The NE–SW-trending Sillon Houiller fault is the main late Variscan fault, which cross-cut the Massif Central along 400 km. Several Late Carboniferous coal basins opened as left-lateral pull-apart basins during faulting. The NE–SW Late Carboniferous to Late Permian extensional tectonics are also responsible for the opening of several half-graben basins upon metamorphic basement, such as Autun, St. Affrique, Lodève, Brive, Rodez, etc. (Fig. 1b). These basins have essentially been filled by continental sediments under tropical climate. Moreover, after the Hercynian orogeny, some of the late Hercynian faults have been reactivated in Mesozoic and Cenozoic times during the Pyrenean orogeny.

Several palaeomagnetic studies have been already carried out in the St. Affrique and Lodève basins. But both basins are located in the southern Massif Central and on the east side of the Sillon Houiller (Mérabet and Guillaume, 1988; Cogné et al., 1990, 1993; Diego-Orozco and Henry, 1993). A new palaeomagnetic study has been carried out from the Permian Brive basin, in the west of the Sillon Houiller (Fig. 1a), in order to (1) recognize the continental deformation of Massif Central after the collision phase, (2) to obtain a more general kinematic view on Massif Central since Permian, and (3) to enlarge the Permian palaeomagnetic database of western Europe. The comparison of our new results with available data from other Permian basins of the Massif Central and from stable northern Europe allows us to test the relative block displacement of the Brive basin with respect to other Permian basins within the Massif Central and to assess the activity of the Sillon Houiller.

2. Geological setting and sampling

The Brive basin is located on the southwestern border of the French Massif Central (Fig. 1b). This NW–SE-elongated Carboniferous–Permian basin of about 600 km^2 is unconformably covered by Triassic–Jurassic formations. The northeastern border is often a normal fault; however, the primary contact is a stratigraphic one, since Carboniferous and Permian rocks unconformably overlie the Limousin metamorphic rocks with basal conglomerate and coarse sandstone. The southern border corresponds to a series of NW–SE-trending faults, namely from west to east: the Châtre, Condat and Meyssac faults. Indeed, these faults do not represent the true margin of the basin, since drillings have shown that Permian rocks extended farther southward below the Mesozoic strata. This fact is in agreement with sedimentological studies that show a southwestward thickening of the deposits (Feys et al., 1979; Feys, 1989). The Meyssac and Condat faults belong to the same NW–SE-trending late Hercynian structure left-laterally offset by the Larche fault and to a lesser extend the Lissac fault (Fig. 1a). Although both NW–SE- and N–S-trending faults were reactivated during the Tertiary Pyrenean compression, earlier motions during late Palaeozoic and Jurassic times control the sedimentation. Dealing with the Carboniferous–Permian events, the Brive basin exhibits a southward-deepening asymmetric shape (Fig. 1c). Such a geometry is in agreement with the half-graben structure found in most of late Palaeozoic basins (e.g., Faure, 1995). Although dynamics of basin opening are not settled, an extentional setting appears likely. For instance, it can be shown that the Chatres horst already existed in the Permian (Feys, 1989).

The sedimentary history of the basin began with the opening of the half-graben during the Late Carboniferous and continues through the Permian without any significant discontinuity. Alike other late Palaeozoic basins of the Massif Central, the age of the continental sediments is poorly documented (Broutin et al., 1986; Feys, 1989). The age of the late Palaeozoic continental formations is determined by facies correlation with other basins. Plant fossils such as *Callipertis conferta* and *Lebachia piniformis* are rare and present a too large stratigraphic range (Mouret, 1891; Odin et al., 1986; Feys, 1989). The Late Carboniferous (Stephanian) sediments consist of grey to black sandstone and conglomerate. Permian sediments of more than 500 m thick are dominated by sandstone–siltite–argilite alternations. The lower part of the Lower Permian is composed of coarse yellowish and grey sandstone and the upper part is marked by the irregular alternation of solid red sandstones and red clay. The Upper Permian (so-called Saxo-Thuringian) rocks are dominated by red sandstones with beds of regular thickness and variable hardness. The transition from Permian to Tri-

Table 1
Sample information and statistic results from higher temperature components

Site	Coordinates	Age	Bedding		n/N	Directions								Poles			
			strike	dip		before bedding correction				after bedding correction				before bedding corr.		after bedding corr.	
						Dec.	Inc.	κ	α_{95}	Dec.	Inc.	κ	α_{95}	λ	ϕ	λ	ϕ
01	45°08′N; 1°27′E	r_{1d}, Autunian	45–70	10–30	8/8	187.6	0.0	41.8	8.7	188.7	−12.7	46.1	8.2	44.4	170.8	50.6	167.8
02	45°05′12″N; 1°25′48″E	r_{2-3c}, Saxo-Thuringian	162	6	7/8	196.7	−18.1	170.5	4.6	194.9	−21.5	170.5	4.6	51.4	154.4	53.7	156.2
03	45°05′N; 1°25′50″E	r_{2-3b}, Saxo-Thuringian	200–210	20–40	6/8	202.9	−4.3	30.1	12.6	200.0	−4.3	38.9	11.1	42.6	149.5	43.6	153.3
04	45°4′30″N; 1°35′51″E	r_{2-3a}, Saxo-Thuringian	240	48	6/10	205.4	−28.1	36.2	11.3	196.7	3.3	36.2	11.3	53.0	138.0	41.0	159.2
05	45°03′N; 1°39′E	r_{2-3a}, Saxo-Thuringian	303	20	6/6	191.8	−26.4	70.6	8.0	193.9	−7.5	70.6	8.0	57.3	160.1	47.0	161.1
06	45°03′30″N; 1°37′E	r_{2-3a}, Saxo-Thuringian	20–30	4–10	8/9	182.4	−2.9	61.7	7.1	182.7	−4.8	29.4	10.4	46.3	178.1	47.3	177.6
07[a]	45°03′15″N; 1°41′E	r_{2-3a}, Saxo-Thuringian	120–125	50–56	7/7	221.9	−22.7	8.3	22.3	209.6	−11.5	48.8	8.7	41.3	121.3	43.1	139.4
			240–250	30–45													
07	45°03′15″N; 1°41′E	r_{2-3a}, Saxo-Thuringian	120–125	50–56	7/7	221.9	−22.7	8.3	22.3	191.4	−11.5	48.8	8.7	41.3	121.3	49.5	164.1
			240–250	30–45													
08	45°03′N; 1°43′E	r_{2-3a}, Saxo-Thuringian	280–290	20–54	6/10	196.3	−30.4	58.6	8.8	195.9	−5.6	65.2	8.4	58.2	151.0	45.5	158.7
09	45°05′N; 1°36′E	r_{2-3b}, Saxo-Thuringian	335–344	12–18	6/6	187.2	−27.9	166.7	5.2	193.4	−19.3	182.8	5.0	59.1	167.9	52.9	158.8
10	45°05′06″N; 1°36′30″E	r_{2-3c}, Saxo-Thuringian	220–250	10–18	6/6	194.7	−22.0	163.1	5.3	191.3	−16.4	144.4	5.6	54.1	156.6	52.0	163.3
11	45°10′N; 1°30′30″E	r_{1d}, Autunian	186	10	6/8	192.7	−2.0	141.0	5.7	192.3	−3.0	141.0	5.7	44.4	163.6	45.0	164.0
12	45°14′20″N; 1°18′30″E	r_{1d} Autunian	111	3	6/6	193.5	−12.7	153.0	5.4	193.4	−15.7	153.0	5.4	49.5	160.6	51.0	160.2
13	45°14′50″N; 1°17′30″E	r_{2-3}, Saxo-Thuringian	116	11	6/6	192.5	−8.4	171.2	5.1	191.9	−18.6	171.2	5.1	47.5	162.6	52.9	161.6
14	45°14′30″N; 1°10′30″E	r_{2-3}, Saxo-Thuringian	Horizontal	0	6/8	189.2	−15.0	175.2	5.1	189.2	−15.0	175.2	5.1	51.6	166.4	51.6	166.4
15	45°14′N; 1°9′24″E	r_{2-3}, Saxo-Thuringian	12–25	10–30	5/8	187.8	−3.6	18.5	18.3	190.0	−6.1	19.0	18.0	46.0	169.9	46.9	166.4
Mean																	
Total					15	194.6	−15.1	32.9	6.8	192.4	−10.6	97.9	3.9	50.6	158.1	48.8	162.5
														$\kappa = 54.1$	$A_{95} = 5.2$	$\kappa = 223.4$	$A_{95} = 2.6$
Autunian					3	191.2	−4.9	115.8	11.5	191.5	−10.5	131.9	10.8	46.1	165.1	48.9	164.0
														$\kappa = 305.5$	$A_{95} = 7.1$	$\kappa = 380.7$	$A_{95} = 6.2$
Saxo-Thuringian					12	195.5	−18.8	32.1	7.8	192.6	−10.6	86.3	4.7	51.7	156.1	48.8	162.2
														$\kappa = 48.2$	$A_{95} = 6.3$	$\kappa = 193.3$	$A_{95} = 3.1$

n/N: number of entry in the statistics/number of treated specimens.
Dec., Inc., κ and α_{95}: declination, inclination, precision parameter and confidence interval at the 95% level.
λ, ϕ: latitude and longitude of polar coordinates.
[a] The result of Site 07 without the inclined fold axis correction and excluded from the statistics.

assic is marked by coarse light-coloured sandstones and conglomerates. The dip of the strata is variable from horizontal up to 60°; even some steeper folding can be observed at the site-scale (about 30 m laterally, e.g. Site 07; see Table 1), which provides a possibility for the fold test.

A palaeomagnetic sampling was carried out in the spring of 1994 with a portable gasoline drilling machine. Fifteen sites have been collected, covering almost all the surface of the basin (more than 50 km in the E–W direction, Fig. 1a). Three sites are located in the upper part of the Lower Permian and twelve in the Upper Permian (Table 1). For each site, eight to twelve cores have been drilled covering a stratigraphic thickness over 10 m. Both magnetic and sun compasses have been used, when it was possible, to orient the drilling cores, and a mean difference of 3° has been identified between the two measurements. This value of the magnetic declination has been used to correct the orientations of the samples and bedding measured only by magnetic compass.

3. Laboratory measurements and data analyses

The palaeomagnetic collection has been treated in the magnetically shielded palaeomagnetic laboratory of the Institut de Physique du Globe de Paris. The cores were cut into specimens of 2.5 cm long, and the lower part of cores has been selected for magnetic measurements. The analysis of the progressive isothermal remanent magnetization (IRM) acquisition experiments have been performed at least in one specimen for each site with a laboratory-built electromagnet and measured with a JR5 spinner magnetometer. Two kinds of magnetic character are found from these rocks: (a) hard coercivity without any magnetic saturation up to 1.2 tesla (Fig. 2a), and (b) coexistence of both soft and hard coercivities with partial magnetic saturation (at about 150 mT; Fig. 2b). Thermomagnetic measurements in free air have also been carried out with a Czech CS3 kapabridge in the Rock Magnetism Laboratory of Orléans, and partial results have been presented in Fig. 2c and d. The bulk magnetic susceptibility (κ) is rather low (in the order of 10^{-6} SI) and close to the sensibility of the apparatus. However, several slope changes of κ versus temperature have been identified (Fig. 2c and d), indicating the composition of different types of magnetic minerals. It seems clear that there the first drop-down is between room temperature and about 120°C, one begins at about 320°C, the third varies from 530 to 550°C and the last is from about 620°C until 680°C, showing the probable existence of goethite, pyrrhotite, (titano)magnetite and hematite. Because of the efficiency of the thermal demagnetization with respect to the alternative field, all specimens have been thermally demagnetized by about fifteen steps in laboratory-built furnaces and measured with a CTF 3-axis cryogenic magnetometer. Demagnetization curves indicate two kinds of magnetic behaviour: (a) the lower (NRM to ~350°C) and higher (350°C to 580°C or up to 680°C) temperature components (LTC and HTC) have similar or very close directions (Fig. 2e and f), or (b) these two components show different directions (Fig. 2g and h). Magnetic directions have been extracted by principal component analysis (Kirschvink, 1980), and site- and age-mean directions were computed using the statistics of Fisher (1953). Bulk magnetic susceptibility was also measured after each heating step to monitor possible change in magnetic minerals. No significant change has been observed for most of the samples. The average of the LTC from about 80% specimens presents a mean magnetic direction of $D_g = 359.6°$, $I_g = 51.2°$ $k_g = 15.7$, $\alpha_{95} = 4.3°$ and $n = 74$ with a better grouping before than after bedding corrections ($D_s = 359.2°$, $I_s = 40.1°$, $k_s = 6.1$ and $\alpha_{95} = 7.3°$), indicating that this component is acquired after the folding and probably recently. The magnetic carriers for this component are probably goethite and pyrrhotite. Most of the specimens show a stable HTC with an unblocking temperature from 350 to 400°C (Fig. 2e–h), up to 680°C in fourteen out of fifteen sites indicating the hematite as the main magnetic carrier (Fig. 2e, f and h). Only one site has an unblocking temperature of 580°C with probably magnetite or titanomagnetite as magnetic carrier (Fig. 2g). Moreover, the magnetic directions isolated from different magnetic carriers, such as magnetites and hematites, are not distinguishable. A reversed magnetic polarity has been isolated for all specimens (Fig. 2e–h). The site-mean directions have been computed (Table 1) and a better grouping has been observed after bedding corrections, even at site-level for seven out of eight sites which present variable dips within the site (Table 1 and Fig. 3). From Fig. 3 and Table 1, it is easy to no-

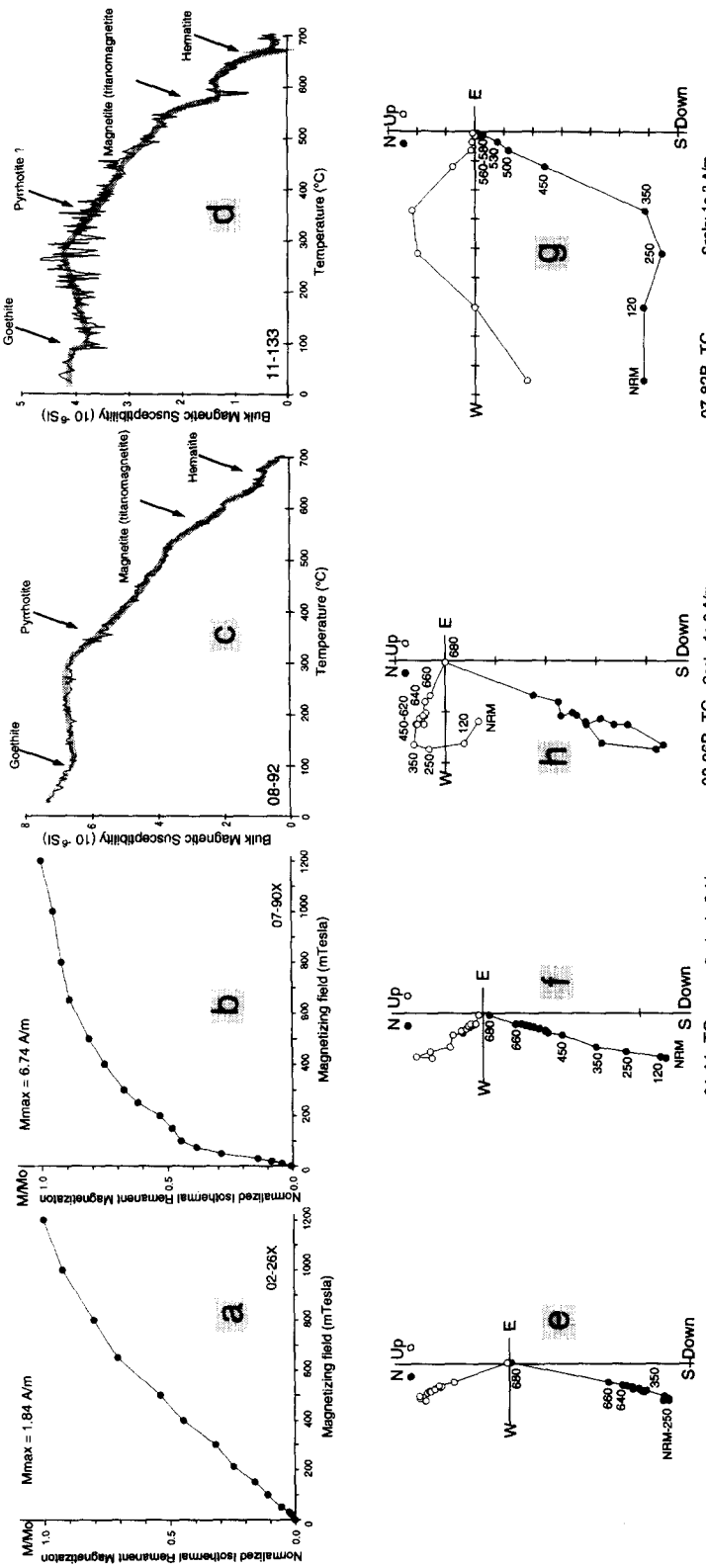


Fig. 2. Results of magnetic study of representative Permian samples from the Brive basin. (a, b) Normalized isothermal remanent magnetization. (c d) Thermomagnetic curves in air. Raw data of measurement are shown by the fine dark lines and the qualitative magnetic mineral interpretation by the shadowed line with arrows. (e–h) Orthogonal vector plots with heating value in degrees Celsius. Solid symbols represent projection in the horizontal plane, open symbols the projection in the vertical plane.

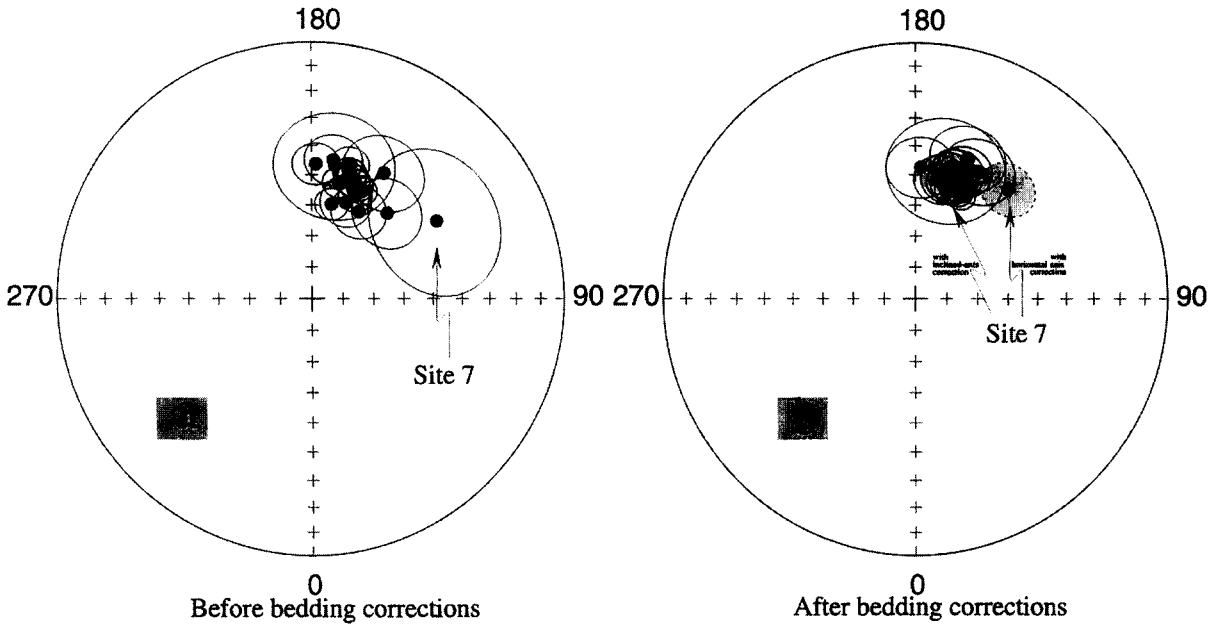


Fig. 3. Equal area plots of 15 site-mean virtual geomagnetic poles with the 95% confidence ovals before (a) and after (b) bedding corrections. Shadowed and dashed (solid) circle = mean-site pole for Site 07 before (after) the inclined folding axis correction.

tice that Site 07 has a better direction distribution after bedding correction. However, if the unfolding procedure is done using the classic method considering that the folding axis is horizontal, the site-mean direction of Site 07 is quite far away from other site-mean directions. The field measurement of the bedding allows us to recognize that the fold axis is inclined by about 23° with respect to the horizontal. Taking into account the correction for such an inclined axis, the site-mean direction for Site 07 falls into the cluster of other sites and gives a better precision for age-mean results. Both age-mean directions and poles have been calculated; the results from the Upper and Lower Permian are in good agreement with an angular difference of $1.2 \pm 6.9°$. Therefore, a single age-mean pole of the Permian has been recalculated (Table 1), which reveals also a significant improvement of distribution after bedding corrections than before at the 99% confidence level ($k_s/k_g = 7.08 > 6.30$, the threshold after McFadden, 1990).

4. Discussion

The laboratory analyses of the field collection has shown the well behaved palaeomagnetic components with universally reversed magnetic polarity for HTC, indicating that the age of the rocks is within the Long Reversal Superchron (Kiaman chron from 320 to 250 Ma). The statistic results present palaeopoles much better grouped after than before folding corrections with the significance at the 99% confidence level. The folding likely occurred before the Jurassic since these strata unconformably cover the Permian rocks. Moreover, the behaviour of two magnetic carriers (magnetite and hematite) for HTC showing an identical magnetic direction assume that there was no important continental drift during the sedimentation and diagenesis periods and that the rock age is representative for the magnetization age. In other words, we can consider that the magnetization carried by the HTC is probably primary and has the age of the rocks.

The results of this study show that the so-called 'Autunian' and 'Saxo-Thuringian' formations do not show any statistical difference in direction ($1.2 \pm 6.9°$). Two hypotheses could be suggested. (1) No palaeomagnetically detectable movement happened during this period from Autunian to Thuringian times. However, the apparent polar wander path for Stable Europe (APWP) shows a sig-

nificant movement from Early Permian (P_E) to Late Permian (P_L) with an angular difference of $4.5 \pm 4.3°$ (Van der Voo, 1990). (2) Although it is clear that Sites 01, 11 and 12 are stratigraphically lower than other sites, the geological duration for the sediments may be less important than attributed by some authors (Feys et al., 1979; Feys, 1989) because the age determination is mainly based on the stratigraphic correlations (colours, facies etc.) instead of characteristic fossils. We therefore suggest that the sampled formations could be of the same geological stage. It will be useful to make more detailed studies on this problem of age determination.

The absolute movement of the Brive basin with respect to the geographic pole is indisputable (about $40 \pm 3°$ in latitude and $3 \pm 3°$ in rotation). However, what is interesting and also the purpose of this study is to recognize the relative movement of this basin with respect to Stable Europe and other surrounding basins.

To compare with Stable Europe, we used Van der Voo's APWP as the reference (Van der Voo, 1990). Based on Bullard's fitting program for the continental reconstruction in the North Atlantic (Bullard et al., 1965), Van der Voo (1990) has combined the mean palaeomagnetic poles from Europe and North America (NA) to establish an APWP for Stable Europe which is partly presented in Fig. 4. A total of 33 poles (20 from Europe and 13 from NA) for the Early Permian (P_E) and 9 poles (5 from Europe and 4 from NA) for the Late Permian (P_L) has been used in this combination and the path is statistically well defined. The palaeopole resulting from this study is closer to the P_L palaeopole (246~266 Ma; $2.3 \pm 4.7°$) than to the P_E one (267~281 Ma; $3.8 \pm 3.2°$; Fig. 4). This observation suggests that the age of our palaeomagnetic collection is probably Late Permian, and, meanwhile, demonstrates that the Brive basin is part of Stable Europe at least since that time. The compatibility of the palaeopoles from the Brive basin and Stable Europe is shown by the non-significant angular difference between them ($2.3 \pm 4.7°$; Table 2 and Fig. 5b and c).

Fig. 5 presents a comparison of the results from the Brive basin with those from previously studied basins including Lodève and St. Affrique which are all situated in the southern Massif Central and in the east of the Sillon Houiller (Table 2; Mérabet and

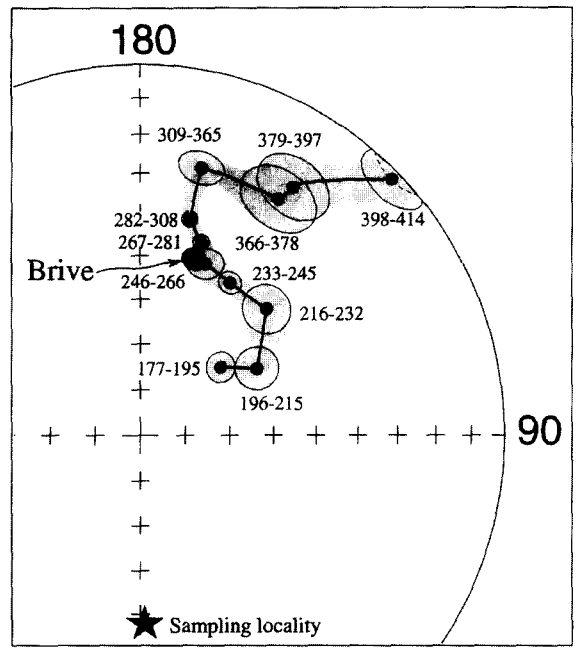


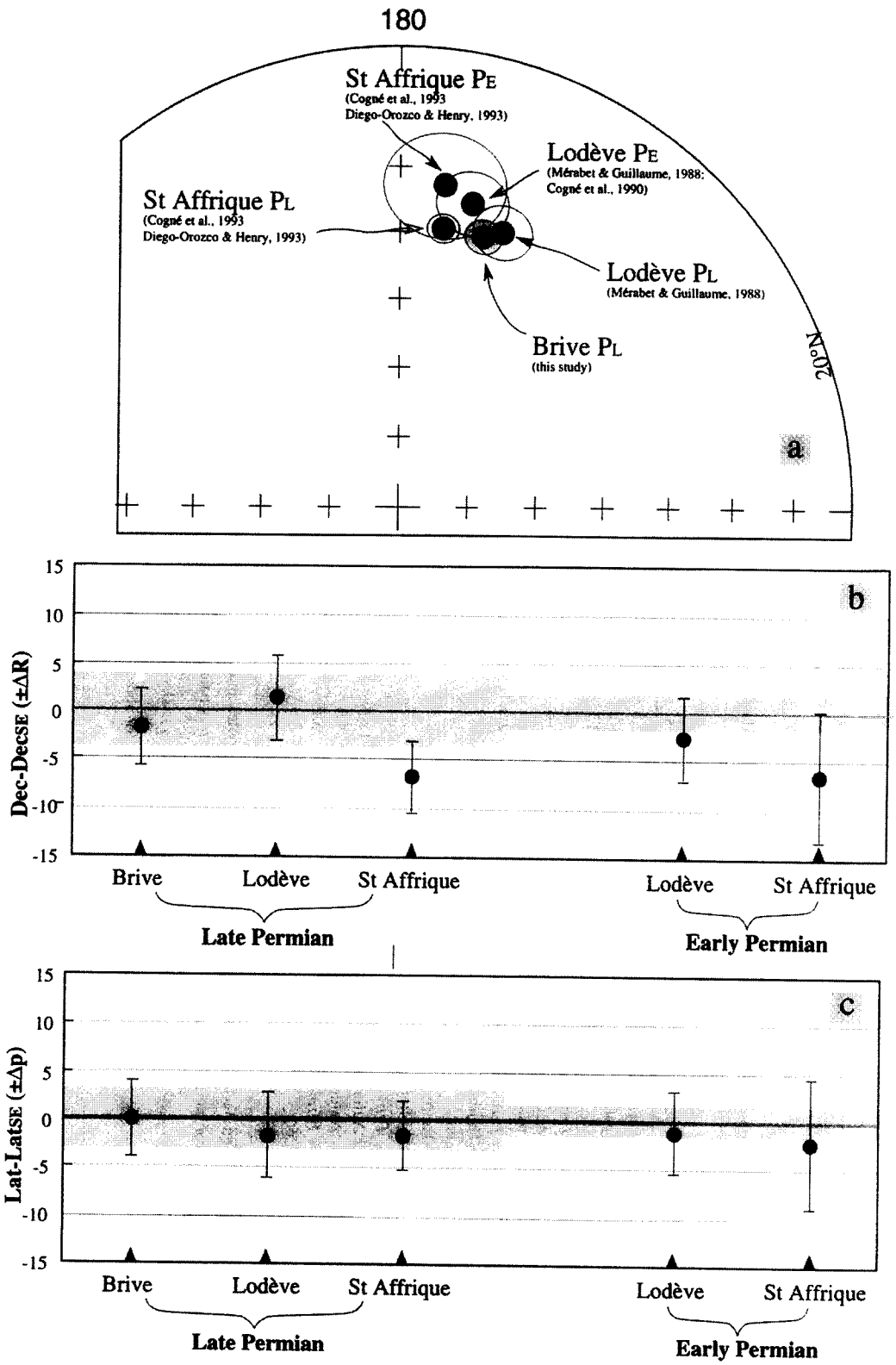
Fig. 4. Equal area projection of the apparent polar wander poles (dots) of Stable Europe (Van der Voo, 1990) and of the palaeomagnetic poles from the Brive basin (triangle) with the 95% confidence ovals. Star shows the sampling locality.

Guillaume, 1988; Cogné et al., 1990, 1993; Diego-Orozco and Henry, 1993). From the palaeopole point of view, the either P_E or P_L palaeopoles are, in first order, compatible without significant differences (Fig. 5a). The minor differences are interpreted in terms of rotation and latitudinal movement (Fig. 5b and c), the difference in latitude being almost negligible (the most important difference among of $2.1 \pm 6.7°$ is between St. Affrique and Stable Europe; Table 2). The difference in rotation is more obvious, but variable for each basin. The relative rotations of Lodève and Brive basins with respect to Stable Europe are not significant ($2.9 \pm 4.2°$ and $2.2 \pm 3.7°$, respectively). Nevertheless, St. Affrique shows a quite important rotation with respect to Stable Europe for both P_E ($6.8 \pm 3.6°$) and P_L ($6.5 \pm 6.7°$). Different mechanisms for this rotation have been proposed by Cogné et al. (1990, 1993) and Diego-Orozco and Henry (1993). Based on the relatively large-scale sampling both in age and space, Cogné et al. (1990, 1993) have obtained progressive counter-clockwise rotations from the bottom

Table 2
Comparison of the relative movements between Permian basins of the French Massif Central and Stable Europe

Basin or plate	Age	Pole				D (ΔD)	λ	$\lambda - \lambda_{SE}$ (ΔP)	$D - D_{SE}$ (ΔR)	Reference
		latitude	longitude	N	A_{95}					
Brive	Upper Permian	48.8	162.5	15	2.6	192.4 (2.6)	5.5	−0.7 (3.7)	−2.2 (3.7)	This study
Lodève	Thuringian	48.6	153.5	6	1.2					Mérabet and Guillaume, 1988
Lodève	Saxonian	43.7	169.5	3	3.1					Mérabet and Guillaume, 1988
Lodève	*Upper Permian*	*47.2*	*159.1*	*9*	*4.1*	*195.0 (4.1)*	*4.5*	*−1.7 (4.5)*	*1.3 (4.5)*	*Average*
Lodève	Upper Autunian	41.3	169.3	5	2.9					Mérabet and Guillaume, 1988
Lodève	Lower Permian	49.4	160.4	3	11.4					Cogné et al., 1990
Lodève	*Lower Permian*	*44.4*	*166.3*	*8*	*4.9*	*190.8 (4.9)*	*0.6*	*−1.2 (4.2)*	*−2.9 (4.2)*	*Average*
Saint Affrique	Saxo-Thuringian	49.0	164.9	8	5.9					Cogné et al., 1993
Saint Affrique	Saxonian	46.9	179.1	7	1.7					Diego-Orozco and Henry, 1993
Saint Affrique	Saxo-Thuringian	49.2	177.3	8	2.0					Diego-Orozco and Henry, 1993
Saint Affrique	Middle Thuringian	49.8	163.4	8	2.8					Diego-Orozco and Henry, 1993
Saint Affrique	*Upper Permian*	*49.0*	*171.0*	*31*	*2.2*	*186.9 (2.2)*	*4.6*	*−1.6 (3.6)*	*−6.8 (3.6)*	*Average*
Saint Affrique	Upper Autunian	41.4	192.8	3	11.5					Cogné et al., 1993
Saint Affrique	Upper Autunian	41.4	161.4	6	4.0					Diego-Orozco and Henry, 1993
Saint Affrique	*Lower Permian*	*42.4*	*171.9*	*9*	*8.2*	*187.2 (8.2)*	*3.9*	*−2.1 (6.7)*	*−6.5 (6.7)*	*Average*
Stable Europe	246–266	49	159	11	3.9	194.6 (3.9)	6.2			Van der Voo, 1990
Stable Europe	267–281	45	162	35	1.9	193.7 (1.9)	1.8			Van der Voo, 1990

Latitude, longitude, N, and A_{95}: paleomagnetic pole coordinates, number of poles and the angle of the cone of confidence.
$D(\Delta D)$: declination (error bar calculated with $\Delta D = \sin^{-1}\left(\frac{\sin A_{95}}{\sin p}\right)$, where p is palaeocolatitude); λ: palaeolatitude; $\lambda - \lambda_{SE}$: difference in latitude of different basins with respect to Stable Europe (error bar calculated with $\Delta p = 0.8\sqrt{A_{95}^2 + (A_{95})_{SE}^2}$); $D - D_{SE}$: difference in declination of different basins with respect to Stable Europe (error bar calculated with $\Delta R = 0.8\sqrt{\Delta D^2 + \Delta D_{SE}^2}$). The geographic coordinates for above calculations are 45.2°N and 1.5°E.


180
St Affrique PE
(Cogné et al., 1993
Diego-Orozco & Henry, 1993)
Lodève PE
(Mérabet & Guillaume, 1988;
Cogné et al., 1990)
St Affrique PL
(Cogné et al., 1993
Diego-Orozco & Henry, 1993)
Lodève PL
(Mérabet & Guillaume, 1988)
Brive PL
(this study)
20°N
a
b
Dec-DecSE (±ΔR)
15
10
5
0
-5
-10
-15
Brive
Lodève
St Affrique
Lodève
St Affrique
Late Permian
Early Permian
c
Lat-LatSE (±Δp)
15
10
5
0
-5
-10
-15
Brive
Lodève
St Affrique
Lodève
St Affrique
Late Permian
Early Permian

to the top of the Permian formation (Autunian to Thuringian). These authors proposed a domino-like kinematic model of the basement to account for rotations at the basin-scale during the formation of the basin from Stephanian to Late Permian times (Van Den Driessche and Brun, 1989). They attributed this deformation to the local variation of the extension directions from ENE–WSW to N–S throughout the Late Carboniferous to Permian in the studied area, in agreement with structural data (e.g., Brunel, 1972; Donnot and Guérangé, 1978; Van Den Driessche and Brun, 1989). On the basis of the independent observations with negligible difference in rotation between P_E and P_L for the St. Affrique basin, Diego-Orozco and Henry (1993) have argued that the rotation of the whole basin with respect to Lodève and Stable Europe should be younger than Thuringian. Both hypotheses suggest that the rotation of St. Affrique is a local phenomenon rather than at the scale of the Massif Central, and it does relate to the activity along the Sillon Houiller. Moreover, the good compatibility of the palaeopoles form the Lodève and Brive basins situated on both sides of Sillon Houiller fault provides an additional evidence for this conclusion. In the southwest of the Massif Central (France), the post-Permian deformation is weak or below the resolution limit of palaeomagnetism.

5. Conclusion

This study of the red sandstones of the Brive basin (Massif Central, France) provides a new reliable Upper Permian palaeomagnetic pole. The comparison of this result with those already available for Stable Europe and other Permian basins in Massif Central indicates that:

(1) The Brive basin has been integrated to Stable Europe since at least the Late Permian, and there is no palaeomagnetically detectable movement of this basin with respect to Stable Europe since this time.

(2) There is no significant movement along the Sillon Houiller fault since Permian time. The relative latitudinal movement among Permian basins of the Massif Central is negligible. No important rotation of these basins has been observed with respect to Stable Europe, except for the St. Affrique basin.

(3) The St. Affrique basin experienced a counterclockwise rotation of ~7° with respect to Stable Europe (Cogné et al., 1993; Diego-Orozco and Henry, 1993), which may be a local phenomenon.

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Fig. 5. Equal area projection of available Permian palaeomagnetic poles from basins of the Massif Central (a), P_L = Late Permian and P_E = Early Permian. Declination (b) and latitudinal (c) differences between Stable Europe, with error bars shadowed, and each basin.

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